Our work to determine appropriate song recommendations for listeners of Spotify through querying the library of music began with testing a few different methods for assessing songs to see if they are about similar topics. These similarity measures included dot product, centering and covariance, correlation, and cosine similarity. From running these methods on a single query song, as well as on all the Top 40 songs by genre, we saw that there were some that outperformed others. Raw dot product appeared to perform the worst in matching similar songs as there was no accounting for song size in this similarity calculation. This means that songs that were longer, and had a higher word mass, were matching many songs just because they had a high volume of common words. To solve this, we needed to find ways to account for song length such that long words were being rewarded less and short songs were being penalized less. We saw that the correlation method performed better in this way because long, non-similar songs started to move down in the ranking while shorter, more similar songs moved up. This makes sense with how correlation is calculated where by rescaling each vector by its standard deviation we are rewarding length of song less. We also saw that cosine similarity also performed better at matching similar songs, and similarly to correlation, had more similar genres with the matched songs to the query songs. By dividing the length of the song vectors, the length of the song is less rewarded in this method. But, through testing all these methods it still appeared that there was room to improve the matching, and particularly that there may be some impact of a high volume of generic words across songs making our matches not as accurate.

Our work to explore the TF-IDF matrix was to try to account for this high-volume of generic words. Being able to weigh words that are extremely common lower than words that are rarer would be important in getting an accurate match. This is because those rare words are more likely to be specific to a genre and more helpful in matching. The TF matrix allows for repeated uses of words to count less and less as the raw count increases. With the IDF vector rare words get high values and more frequent words get low values such that when broadcasted onto TF gives a reweighted matrix that will place less value on high frequency words in the matching process. When we used the TF-IDF matrix in our dot product, covariance, correlation, and cosine similarity methods, we saw improvements in matching. This included more top matches being similar genres and having titles with similar sentiments.

To test this improvement further, we used a precision calculation to look at which method was able to have the most genre matches between the query song and the top ranked similar songs from the method across the Top 40 by genre. We looked at comparing raw dot product, raw cosine similarity, and TF-IDF cosine similarity. Overall, we saw that raw dot product had a precision score, or average number of matching genres per query song, of 0.09. For raw cosine similarity, the precision score was 0.19, and for TF-IDF cosine it was 0.210. Overall, TF-IDF cosine performed the best out of these methods, but there was still room for improvement.

My idea for a new system to see if I could get a better precision score was removing the most frequent words altogether rather than just reweighting them. Because these more frequent words don't seem to provide anything specific to any songs, and therefore are unhelpful with matching, I thought that there would be minimal consequence in removing them and that it could ultimately improve the score. To do so, I sorted the vector of all words by their IDF weight and filtered the top 50 out of the song-by-word matrix. I then re-did the TF-IDF cosine method using this filtered song-by-word matrix. I chose to only test the TF-IDF cosine method as that was the most precise method we had found so far. I found that my system outperformed TF-IDF cosine only very slightly with a precision score of 0.213 which is 0.003 greater than that of TF-IDF cosine.